

Zichen Zhu

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EDUCATION

Vanderbilt University

B.S. in Computer Science and Applied Mathematics

Nashville, TN, USA

Aug. 2023 – May. 2027 (Expected)

- **GPA:** 3.986/4.000
- Tau Beta Pi Engineering Honor Society: Inducted as top 12.5% of the Class of 2027
- Selected coursework: Foundations of Machine Learning, Applied Machine Learning, Intermediate Software Design, Algorithm, Artificial Intelligence, Linear Algebra, Probability and Statistics for Engineering

WORK EXPERIENCE

Institute for Software Integrated Systems, Vanderbilt University

Undergraduate Research Assistant

Nashville, TN, USA

May. 2026 - Present

- Built an exact robustness verifier for a trained convolutional first-to-fire spiking neural network, implementing star-set reachability and property-directed branch-and-bound.
- Designed a property-directed pruning strategy that confines branching to firing decisions capable of realizing a pending violation, reducing search nodes, LP queries, and runtime by 8–26× with provably identical verdicts.
- Benchmarked 20 MNIST queries across six perturbation radii and two threat-model sizes, delivering complete verdicts through $\varepsilon = 0.10$ ($k = 1$) and $\varepsilon = 0.07$ ($k = 2$); presented results in a poster and technical talk.

Department of Computer Science, Vanderbilt University

Teaching Assistant and Grader for CS3251/5251 & CS1101

Nashville, TN, USA

Aug. 2024 - Present

- Conducted weekly office hours (3+ hours per week) to assist students with programming concepts in C++ and Java, debugging, and assignment-related queries.
- Graded programming assignments, midterm exams, and final exams for over 380 students, ensuring accurate and timely feedback to support students' learning progress.

China Telecom AI Company (TeleAI)

Product Development Intern

Beijing, China

Jun. 2024 - Jul. 2024

- Conducted comprehensive market research on competing English-speaking language models products and analyzing their features, performance, and market positioning.
- Authored detailed product and market analysis reports, providing insights and recommendations to support strategic decision-making in product development.
- Gained proficiency in prompt engineering techniques to enhance the performance and accuracy of the company's Xingchen large language model for English-speaking applications, optimizing user interactions and task completion.

PROJECT EXPERIENCE

AlphaZero Reinforcement Learning Pipeline for Connect Four and Backgammon

Python, MCTS, Reinforcement Learning

- Built an AlphaZero-style self-play reinforcement learning system with a dual-headed policy-value network and PUCT-guided Monte Carlo Tree Search, reaching **95%+** win rate against a random Connect Four baseline.
- Proposed and implemented a **Expecti-MCTS** variant that interleaves player decision nodes with probabilistic chance nodes, enabling AlphaZero-style learning under stochastic state transitions.
- Adapted AlphaZero to stochastic Backgammon by implementing Chance-Aware MCTS, explicitly branching over dice-roll chance nodes and outperforming standard MCTS baselines with **75%** win rate.

English Football Player Market Value Prediction

Python, XGboost, MLP, GNN

- Developed a soccer player valuation pipeline to classify player-season observations into market-value tiers using tabular features, categorical embeddings, and temporal season-based splits.

- Modeled club-season teammate relationships as a graph, enabling GraphSAGE message passing to capture relational team-context signals beyond isolated player statistics.
- Compared XGBoost, MLP, MLP + ClubID, and teammate-edge GraphSAGE models, with the GNN achieving the best test performance at **0.580** accuracy and **0.550** macro-F1.

SKILLS

- Programming Language: Java, Python, C/C++, Racket, Prolog
- Language: Mandarin, English